SHADAAB HASAN

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SUMMARY

Data scientist with an M.Sc. in Computer Science (Data Analytics) from the University of Galway (2.1). Hands-on experience in computer vision and NLP: training object detectors, fine-tuning transformer models and evaluating models on fair test sets. Published research on skin lesion classification. Looking for data scientist roles in Ireland.

EDUCATION

M.Sc. Computer Science (Data Analytics) | University of Galway, Ireland 2025 – 2026
Result: 2.1 (Second Class Honours, Grade 1). **Modules:** Machine Learning, Natural Language Processing, Modelling, Statistics for Data Science.

B.E. Computer Engineering | Rizvi College of Engineering (University of Mumbai) 2021 – 2025
CGPA: 7.8 / 10. **Modules:** Machine Learning, Applied Data Science, NLP, Data Structures and Algorithms.

RESEARCH

M.Sc. Thesis | Python, PyTorch, Linux, Git 2026
Applying and Evaluating Vision Transformers for Continuous Multi-Object Tracking of Honeybees using Event Cameras

- Applied and evaluated a pretrained vision transformer point tracker (ETAP, CVPR 2025) on honeybee recordings from Prophesee event cameras, without retraining.
- Designed a three-layer evaluation: a synthetic EventKubric baseline (Average Jaccard 20.8%), the labelled AEMOT BeeSwarm dataset, and my own unlabelled campus recordings.
- Built a precision/recall evaluation harness against published AEMOT results, rebuilding per-frame ground truth with DBSCAN and scoring with Hungarian matching; decoded corrupted EVT3 sensor files to make the data usable.
- Found that tracking drift, not detection, was the bottleneck (median error rose from 3.5 px to over 15 px within ~33 ms); a parameter sweep roughly doubled precision and recall over default settings.

PROJECTS

Lightweight Object Tracking | Group project | Python, YOLO11n, OpenCV 2026

- Trained YOLO11n on Pascal VOC (20 classes), comparing three learning-rate and batch-size settings; the best reached mAP@50 of 0.806.
- Built the OTB benchmark evaluation (success AUC, precision at 20 px, FPS) comparing a Kalman filter motion tracker with OpenCV KCF and CSRT, and co-wrote the IEEE-format report.

Student Feedback Sentiment Analysis | Team project, model lead | Python, Hugging Face, scikit-learn 2025

- Built the model for a teacher feedback app: combined ~72,000 texts (SST-5 reviews, TweetEval tweets and 1,302 student comments) into a 3-class sentiment dataset.
- Compared Naive Bayes, Random Forest, Logistic Regression, XGBoost, an LSTM with GloVe embeddings and fine-tuned DistilBERT/BERT; BERT reached 72% accuracy on 14,612 test texts vs 62% for the best classic model.

Skin Cancer Detection | Team lead | Python, TensorFlow, Keras Feb – May 2024

- Led a team of four building and comparing four CNN architectures to classify skin lesions into 7 types using HAM10000 (10,015 images); results published in JETIR (2024).

Hotel Booking Cancellation Risk | R, caret, randomForest Aug – Dec 2024

- Predicted cancellations on 119,390 bookings: Random Forest reached 0.85 AUC vs 0.75 for a logistic regression baseline; threshold tuning raised cancellation recall from 53% to 74%; partial dependence plots explained the main drivers.

PUBLICATION

"Skin Cancer Recognition System", Journal of Emerging Technologies and Innovative Research (JETIR), June 2024.

SKILLS

Languages: Python, R, SQL
Machine learning: scikit-learn, PyTorch, TensorFlow/Keras, Hugging Face Transformers, XGBoost, YOLO (Ultralytics), OpenCV
Data: Pandas, NumPy, tidyverse, ggplot2, Matplotlib, Seaborn, Excel
Tools: Git, Linux, Jupyter, Weights & Biases, AWS (Cloud Foundations)

CERTIFICATIONS

Applied Machine Learning in Python (University of Michigan, Coursera) | AWS Academy Cloud Foundations | Data Analytics with Python (NPTEL) | High-Performance Leadership (Santander Open Academy)

LEADERSHIP AND OTHER EXPERIENCE

Sports Coordinator, Rizvi College: organised a college football tournament for 16 teams across 4 batches (112 players).
Retail Associate, TK Maxx Galway (Oct 2026 – present, part-time); **Kitchen Porter, Kai Restaurant Galway** (Jul – Sep 2026, part-time).